



GLOWING FUTURE ACADEMY

Primary School Management System

Phase 2 — New Features Documentation



Version 2.0 · June 2026 · React + TypeScript + Supabase

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■ Document	■ Phase	■ Date	■ Status
Phase 2 Features Addition Guide	Completing PSMS Requirements	June 2026	In Progress

1. Current System vs PSMS Requirements

This document tracks the complete comparison between our GFA React web system and the original PSMS Access documentation. It serves as a step-by-step guide for adding all missing features.

Goal: Make the GFA web system match every feature in the PSMS v1.2 documentation while being superior in design, accessibility, and technology.

1.1 What We Have Already Built

Module	Feature	Status
Students	Full registration with all fields	■ Complete
Teachers	Staff management with dynamic subjects	■ Complete
Classes	Grade/section setup with occupancy tracking	■ Complete
Attendance	Daily marking, mark all, upsert logic	■ Complete
Grades	Score recording (single score)	■■ Partial
Fees	Payment recording and tracking	■■ Partial
Library	Full book inventory management	■ Complete
Events	School calendar with types	■ Complete
Reports	Attendance, fees, grades, enrollment	■ Complete
Settings	School profile, academic years, subjects	■ Complete
Parents	Guardian records with student linking	■ Complete
Parent Portal	Separate parent login with child data view	■ Complete
Academic Years	Year management with current year flag	■ Complete

1.2 What We Still Need to Build

Feature	PSMS Reference	Priority
CA1 + CA2 + Exam score breakdown	Section 8, tbl_ExamScores	■ High
Auto-generate Student ID	Section 10.1, GenerateAdmissionNo()	■ High
Student Report Card (printable)	Section 9, rpt_ReportCard	■ High
Fee Receipt (printable)	Section 9, rpt_FeeReceipt	■ Medium
Fee Invoice per student	Section 9, rpt_FeeInvoice	■ Medium
Auto Receipt Number (RCP-YYYY-NNNNN)	Section 10.2	■ Medium
Pickup Log module	Section 3.7, tbl_PickupLog	■ Medium
Emergency Contacts report	Section 9.1, rpt_EmergencyContacts	■ Medium
Class position/ranking	Section 8.3, GetClassPosition()	■ Low
Fee overpayment validation	Section 10.5	■ Low
Attendance below 75% flag	Section 9, rpt_AttendanceSummary	■ Low

2. Phase 2 Build Plan

All missing features grouped into 3 phases for organized development:

Phase 2A — Database Updates	■ Run SQL: Add ca1_score, ca2_score, exam_score columns to grades table ■ Run SQL: Add receipt_number column to fees table ■ Run SQL: Create pickup_log table ■ Update Grades page to accept CA1 + CA2 + Exam entry ■ Update Grades table display to show breakdown
Phase 2B — Auto Generation	■ Auto-generate Student ID (GFA-2025-001 format) ■ Auto-generate Receipt Number (RCP-2025-00001 format) ■ Update Students page to show auto-generated ID ■ Update Fees page to show/generate receipt numbers
Phase 2C — New Modules	■ Build Pickup Log module (log parent pickups) ■ Add Emergency Contacts to Reports ■ Add Attendance below 75% flag to Reports ■ Add Fee Collection Summary to Reports
Phase 2D — Printable Documents	■ Student Report Card (per student, per term) ■ Fee Receipt (printable, RCP number) ■ Fee Invoice per student ■ Class Performance report

3. Detailed Feature Specifications

3.1 CA1 + CA2 + Exam Score Breakdown

The PSMS uses a 3-part grading system per the Access documentation Section 8: CA1 (max 20) + CA2 (max 20) + Exam (max 60) = Total (max 100).

Component	Max Marks	Description
CA1 Score	20 marks	First Continuous Assessment
CA2 Score	20 marks	Second Continuous Assessment
Exam Score	60 marks	End of Term Examination
Total Score	100 marks	CA1 + CA2 + Exam (auto-calculated)
Grade	A / B / C / D / F	Auto-assigned from total score

Grading Scale (from PSMS Section 8.1)

Score Range	Grade	Remark
70 – 100	A	Excellent
60 – 69	B	Very Good
50 – 59	C	Good
40 – 49	D	Pass
0 – 39	F	Fail

■ SQL needed: ALTER TABLE grades ADD COLUMN ca1_score, ca2_score, exam_score

3.2 Auto-Generate Student ID

From PSMS Section 10.1 — Admission numbers should be auto-generated in a consistent format to prevent duplicates and make records easy to find.

Format: GFA-YYYY-NNNN Example: GFA-2025-0001, GFA-2025-0002

How it works:

- When admin clicks '+ Add Student', the system auto-generates the next available ID
- Format uses current year + sequential number
- Admin can override if needed but default is auto-generated

- ID is unique and indexed — no duplicates allowed

3.3 Auto-Generate Fee Receipt Number

From PSMS Section 10.2 — Receipt numbers must be auto-generated in RCP-YYYY-NNNNN format for proper audit trails and year-based tracking.

Format: RCP-YYYY-NNNNN Example: RCP-2025-00001, RCP-2025-00002

3.4 Pickup Log Module

From PSMS Section 3.7 — Every parent pickup must be logged with timestamp for safety and audit purposes. This is a safety-critical feature.

Field	Type	Description
student_id	UUID	Reference to the student being picked up
parent_id	UUID	Reference to the parent/guardian
pickup_datetime	Timestamp	Date and time of pickup
verification_method	Text	Manual / Override
verified_by	Text	Name of staff who confirmed pickup
override_pin_used	Boolean	Emergency override was used
notes	Text	Additional notes

3.5 Student Report Card

From PSMS Section 9, rpt_ReportCard — A printable report card showing a student's academic performance per subject per term. This is one of the most important deliverables for a school system.

The report card will include:

- School header with GFA logo, name and motto
- Student details — name, class, student ID, academic year, term
- Subject-by-subject results — CA1, CA2, Exam, Total, Grade
- Overall position in class
- Attendance summary for the term
- Teacher's remark and principal's signature area
- Print button to generate PDF

3.6 Fee Receipt & Invoice

From PSMS Section 9 — rpt_FeeReceipt and rpt_FeeInvoice. Printable documents given to parents as proof of payment.

Fee Receipt includes:

- Receipt number (RCP-YYYY-NNNNN format)
- School name, date, and payment details
- Student name, class, and student ID
- Fee type, amount paid, and balance
- Print button

Fee Invoice includes:

- All fee items for the student's class and term
- Amount due, amount paid, and outstanding balance
- Due dates for each fee item
- Print button

3.7 Emergency Contacts Report

From PSMS Section 9.1, rpt_EmergencyContacts — A class-by-class printable list of all students with their parent/guardian contact details. Essential for school safety and emergency situations.

Report includes:

- Sorted by class then student name
- Student name and admission number
- Parent/guardian name and relationship
- Primary phone and secondary phone
- Whether parent is authorized for pickup

4. Step-by-Step Implementation Guide

Follow these steps in exact order to add all missing features:

Step 1	Run Database SQL Go to Supabase SQL Editor and run the Phase 2 SQL script. This adds: ca1_score, ca2_score, exam_score to grades table Adds: receipt_number to fees table Creates: pickup_log table
Step 2	Update Grades Page Replace GradesPage.tsx with new version that has: • CA1 + CA2 + Exam input fields • Auto-calculated total score • Updated table display showing all 3 components
Step 3	Auto Student ID Update StudentsPage.tsx to: • Auto-generate GFA-YYYY-NNNN on Add Student • Query last student ID and increment • Allow manual override
Step 4	Auto Receipt Number Update FeesPage.tsx to: • Auto-generate RCP-YYYY-NNNNN on fee record creation • Display receipt number in the fees table
Step 5	Build Pickup Log Create new PickupLogPage.tsx: • Search student by name or ID • Select authorized parent from linked parents • Log pickup with timestamp • Add to Sidebar and Routes
Step 6	Update Reports Add to ReportsPage.tsx: • Emergency Contacts report • Attendance below 75% flag (highlight in red) • Fee Collection Summary • Pickup Log report

Step 7

Student Report Card

Create ReportCard component:

- Accessible from Student profile
- Select term and academic year
- Shows all subjects with CA1+CA2+Exam+Total+Grade
- Print/Download as PDF

Step 8

Fee Receipt & Invoice

Add print functionality to Fees:

- Fee Receipt button on each payment record
- Fee Invoice button per student
- Formatted printable layout with GFA header

5. SQL Scripts

Run these SQL scripts in your Supabase SQL Editor:

5.1 Phase 2A — Database Updates

```
-- Add CA1, CA2, Exam columns to grades table
ALTER TABLE grades ADD COLUMN IF NOT EXISTS ca1_score NUMERIC(5,2) DEFAULT 0;
ALTER TABLE grades ADD COLUMN IF NOT EXISTS ca2_score NUMERIC(5,2) DEFAULT 0;
ALTER TABLE grades ADD COLUMN IF NOT EXISTS exam_score NUMERIC(5,2) DEFAULT 0;

-- Add receipt number to fees table
ALTER TABLE fees ADD COLUMN IF NOT EXISTS receipt_number VARCHAR(20);

-- Create Pickup Log table
CREATE TABLE IF NOT EXISTS pickup_log (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  student_id UUID REFERENCES students(id) ON DELETE CASCADE NOT NULL,
  parent_id UUID REFERENCES parents(id) ON DELETE SET NULL,
  pickup_datetime TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  verification_method VARCHAR(20) DEFAULT 'Manual',
  verified_by VARCHAR(100),
  override_pin_used BOOLEAN DEFAULT false,
  notes TEXT,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Enable RLS
ALTER TABLE pickup_log ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Allow all" ON pickup_log FOR ALL USING (true) WITH CHECK (true);
GRANT ALL ON TABLE pickup_log TO anon, authenticated;
```

5.2 Verify Changes

```
-- Verify grades table has new columns
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'grades'
ORDER BY ordinal_position;

-- Verify pickup_log table exists
SELECT * FROM pickup_log LIMIT 1;

-- Verify fees has receipt_number
SELECT column_name FROM information_schema.columns
WHERE table_name = 'fees' AND column_name = 'receipt_number';
```

6. Files to Create / Update

Complete list of all files that need to be created or modified:

Action	File Path	What Changes
UPDATE	src/pages/grades/GradesPage.tsx	Add CA1+CA2+Exam fields, update display
UPDATE	src/pages/students/StudentsPage.tsx	Auto-generate student ID
UPDATE	src/pages/fees/FeesPage.tsx	Auto-generate receipt number
UPDATE	src/pages/reports/ReportsPage.tsx	Add emergency contacts, 75% flag
CREATE	src/pages/pickup-log/PickupLogPage.tsx	New pickup log module
CREATE	src/components/students/ReportCard.tsx	Printable report card component
CREATE	src/components/fees/FeeReceipt.tsx	Printable fee receipt component
CREATE	src/components/fees/FeeInvoice.tsx	Printable fee invoice component
UPDATE	src/routes/index.tsx	Add pickup-log route
UPDATE	src/components/layout/Sidebar.tsx	Add Pickup Log to nav

7. Implementation Checklist

Use this checklist to track progress as we build each feature:

Database	■ Run Phase 2A SQL script in Supabase ■ Verify ca1_score, ca2_score, exam_score columns in grades table ■ Verify receipt_number column in fees table ■ Verify pickup_log table created successfully
Grades Module	■ Replace GradesPage.tsx with CA1+CA2+Exam version ■ Verify total auto-calculates correctly ■ Verify grade label shows correct letter grade ■ Test adding a grade record with all 3 components
Students Module	■ Update StudentsPage.tsx with auto ID generation ■ Test: add student and verify ID format GFA-2025-0001 ■ Test: add second student and verify ID increments
Fees Module	■ Update FeesPage.tsx with receipt number generation ■ Test: add fee record and verify RCP-2025-00001 format ■ Test: add second fee and verify increment
Pickup Log	■ Create src/pages/pickup-log/ folder ■ Create PickupLogPage.tsx ■ Add route to src/routes/index.tsx ■ Add to Sidebar navigation ■ Test: log a pickup and verify it saves
Reports	■ Add Emergency Contacts report tab ■ Add 75% attendance flag (highlight in red) ■ Add Fee Collection Summary report ■ Test all report types generate correctly
Report Card	■ Create ReportCard component ■ Add Report Card button to student records ■ Test: generate report card for a student ■ Verify all subjects show with CA1+CA2+Exam+Total+Grade ■ Test print functionality
Fee Documents	■ Create FeeReceipt component ■ Create FeeInvoice component ■ Add print buttons to Fees page ■ Test receipt prints with correct receipt number ■ Test invoice shows all fee items for student